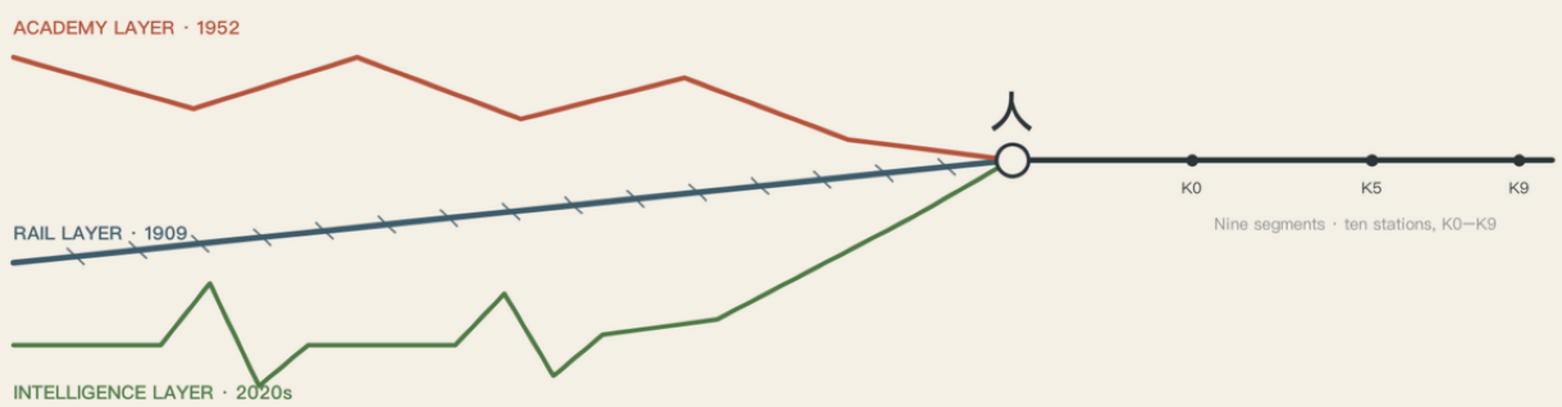
JING–ZHANG AI INNOVATION BELT · URBAN DESIGN OPEN CALL · CONCEPT PROPOSAL

JINGZHANG TRIFOLD

京张三叠

One corridor imprinted with three leaps of national basic capability — mobility network → knowledge network → intelligence network

RAIL 1909 · ACADEMY 1952 · INTELLIGENCE 2020s | One Spine, Two Corridors, Three Harbors · Nine Segments, Ten Stations



Three lines converge at the 「人」 switchback node — reversal as mechanism: sandbox rollback · accessible ramp · failure gallery

Site strip silhouette (vertical · north up)
~9.7 km N–S × ~1.3 km E–W
overall design area 11.41 km² (provisional)

Conceptual reference scheme · provisional | all boundaries are surrogate, NOT an official red line | Sources: site package · public source registry

Jingzhang Trifold

京张三叠 · JINGZHANG TRIFOLD — Rail · Academy · Intelligence Meridian

RAIL LAYER · 1909	ACADEMY LAYER · 1952	INTELLIGENCE MERIDIAN · 2020s

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PROVISIONAL

PROVISIONAL · All spatial recommendations are conceptual reference schemes generated on a registered provisional coarse boundary; they do not replace statutory planning and constitute no government–approved conclusion. Once official boundary and key–area polygons are published, all area metrics must be recomputed.

Concept & Structure · The Trifold Narrative and Spatial Skeleton

Rail Layer · 1909

The first trunk railway designed and built by Chinese engineers departed from here; Zhan Tianyou solved the Qinglongqiao gradient with a herringbone switchback. The Rail Layer built a network of human mobility — the layer of the builders.

Academy Layer · 1952

The 1952 reorganization built the 'Eight Colleges' along Xueyuan Road — with the CAS institutes in Zhongguancun, the republic's first systematic disciplinary cluster. The Academy Layer built a network of knowledge production — the layer of the scholars, and the enabling condition of the Meridian.

Intelligence Meridian · 2020s

The Dazhongsi—Zhichun—Wudaokou line hosts China's densest AI cluster. The Meridian is building a network of intelligent services — built for AI, it must also reserve interfaces for the next general–purpose technologies after AI.

One Spine · Two Corridors · Three Harbors · Nine Segments, Ten Stations

One Spine

The Jing–Zhang Railway Heritage Park main spine — a north—south slow–mobility greenway, centerline approx. 9.6 km, the physical carrier of the Rail Layer and the public–life main axis.

Two Corridors

East: Xueyuan Road—Xiaoyue River composite corridor (learning–vein interface, blue–green ecology and low–risk scenario testing — the Xiaoyue River Scenario Enablement Wing); West: Zhongguancun sci–tech services liaison corridor (factor and capital functions of the Technology Services Wing).

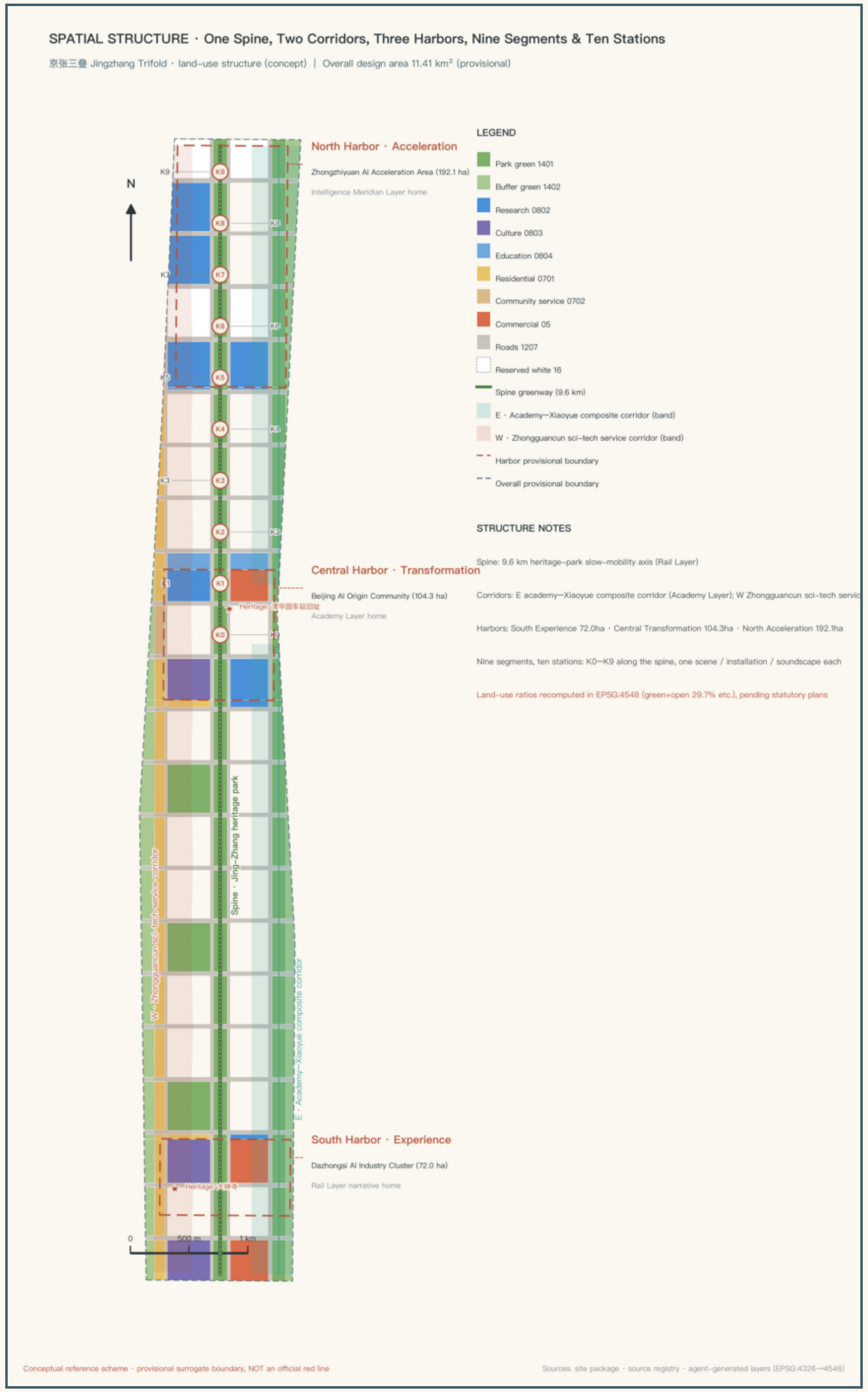
Three Harbors

South Harbor Dazhongsi (experience) / Central Harbor AI Origin Community (conversion) / North Harbor Zhongzhiyuan (acceleration) — narrative home grounds of the three layers; per the Trifold Protocol each flagship node holds historical evidence, public learning, and an exitable AI service at once.

Nine Segments, Ten Stations

Ten station markers K0—K9 along the spine form nine narrative mileage segments; each station is one scenario, one public artwork, one soundscape — a walkable century of history.

Spatial Structure · Three–Level Scope and Land–Use Concept



Three–Level Scope Framework

Coordinated Research Area · approx. 43.6 sq km

North Fifth Ring / Jingzang Expressway / Xizhimen Outer St / Wanquanhe Rd. Industrial strategy and regional coordination only; no spatial siting conclusions.

Overall Design Area · approx. 11.4 sq km

Urban districts 1—2 km around the heritage park; regulatory–plan–depth urban design. Computed area approx. 11.4128 million sq m (approx. 0.1% deviation from the announcement caliber).

Key–Area Detailed Design · approx. 368.4 ha

North Harbor Zhongzhiyuan / Central Harbor AI Origin Community / South Harbor Dazhongsi, each at integrated–implementation–plan depth.

Land–Use Concept Balance (computed · provisional caliber)

Green & open space
≈ 29.7%

Research·Education·Culture
≈ 26.4%

Residential
≈ 15.6%

Roads
≈ 15.2%

Commercial
≈ 8.7%

Reserved blank land
≈ 4.4%

All ratios are computed from the submitted land–use layer under EPSG:4548; the standalone green–layer caliber gives approx. 29.5% coverage — both calibers are disclosed.

Three Harbors Detailed Design · Three Narrative Home Grounds

South Harbor · Dazhongsi AI Industry Cluster (computed approx. 72.0 ha)

Rail Layer home ground · the 'departure hall' of the belt

Positioning: a living room where a century of memory overlaps AI–native consumption. 'One square, one installation, one market' — the urban living–room square connects the Dazhongsi metro hub; the 'New Bell' light–and–sound installation lets the Yongle Great Bell converse with an AI urban soundscape — the belt's only primary landmark and the core of its auditory identity; the open–source market and unmanned–retail test street form an AI–native consumption band. Per the Trifold Protocol it holds historical evidence, public learning, and exitable AI services at once. Risk: complex commercial property rights — negotiated micro–renewal, no large–scale demolition assumptions.

Central Harbor · Beijing AI Origin Community (computed approx. 104.3 ha)

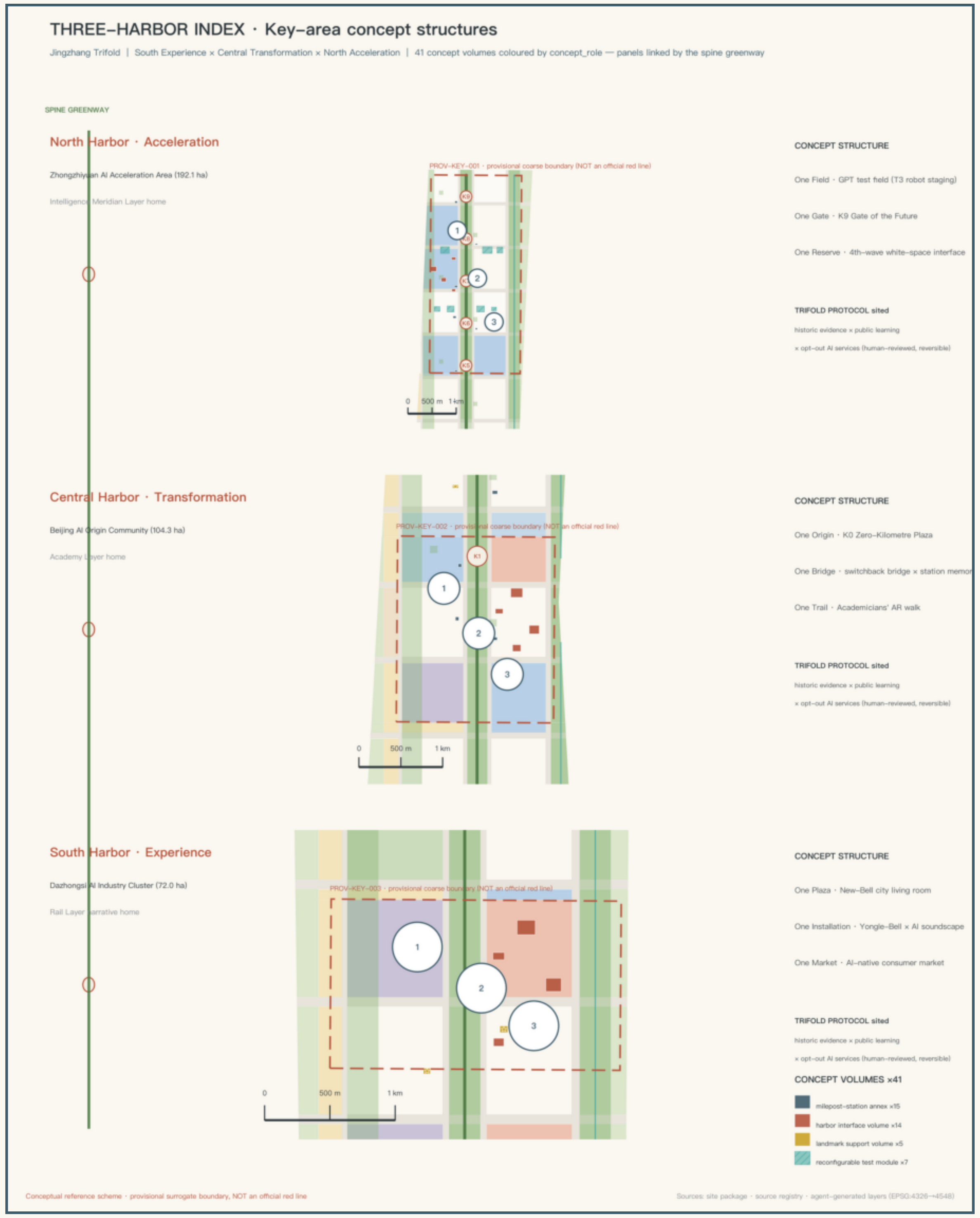
Academy Layer home ground · the 'Kilometer Zero' of the belt

Positioning: the hub converting knowledge into ecosystem, the 'origin' of Chinese AI. 'One origin, one bridge, one trail' — the K0 Origin Square is the conceptual kilometer zero of this belt's walking narrative system (the historical mileage origin of the Beijing—Zhangjiakou Railway lies in Liucun, Fengtai — not on this site), overprinting the memorial meaning of the 'AI Origin'; the Herringbone Switchback Bridge borrows the switchback wisdom to organize accessible ramps and a 'problem—experiment—failure—correction' exhibition; the Academician Trail AR guide tells the stories of the Eight Colleges and Zhongguancun scientists. Risk: multi–party property rights and existing park operations require consultation.

North Harbor · Zhongzhiyuan AI Acceleration Area (computed approx. 192.9 ha)

Intelligence Meridian home ground · the 'acceleration harbor' facing the future

Positioning: spatial carrier of the full–stack independent innovation system and the reserved interface for the next general–purpose technology. 'One field, one gate, one blank' — a reconfigurable Intelligence Meridian test field; the K9 Gate of the Future is a walkable, lingerable, humble threshold signaling 'the next corridor' with low–brightness light pulses at night — not a high–recognition sculptural landmark; approx. 4.4% blank land reserved for the technologies 'after AI'. Risk: construction conditions near the Fifth Ring need verification; blank–land activation rules should be defined by governance.



The three key–area polygons are coarse provisional rectangles; rectangle edges must not be read as parcel lines or road red lines; internal structures are directional expressions only.

Scenario Cards & Personas · AI behind the scenes, people in front

No.	Scenario	Location	Target users
S1	Platform Morning Reading (AI study pavilion)	K1 & Platform Park	Students
S2	Academician Trail AR guide	Learning–vein corridor	Visitors/students
S3	Crossing Coffee · Developers' Living Room	Central Harbor	Developers
S4	Night–Walk Guardian lighting	Entire main spine	Late–returners/residents
S5	Memory–Collection Vehicle	Three harbors rotating	Long–time residents

No.	Scenario	Location	Target users
S6	Silver–Hair Digital Station	South Harbor + communities	Older adults
S7	Open–Air Algorithm Cinema	K4 Platform Park	Citizens
S8	Open–Source Market	South Harbor	Makers/citizens
S9	Rail Concerts	K6 / Herringbone Bridge	Citizens
S10	Accessible Tactile Soundscape	Entire main spine	Visually impaired etc.

AI industry testing & validation scenarios (conceptual testing · stoppable · reversible)

T1 · TEST

Low–speed unmanned delivery test section: a time–windowed zone in the Xiaoyue River corridor (eastern edge) with public testing rules and accident–handling procedures.

T2 · TEST

AI music–generation public test field: carried by the Rail Concerts; every track labeled for generation method and licensing status.

T3 · TEST

Urban service robot staging unit: 'stage first, then go live' at the North Harbor; robots enter open areas only after safety assessment and public notice.

Six User Personas

- Long–time residents · guardians of site memory
- University students · masters of the Academy Layer
- AI researchers & developers · need community and computing power
- International entrepreneurs · one–stop landing services
- Families & visitors · safe, fun, educational experiences
- Late–night commuters · a lit, humane way home

Flagship scenarios in depth (full mechanism design · others remain supporting cards)

S4 Night–Walk Guardian

Inputs: anonymous crowd density and illuminance only; no facial recognition; on–device processing with 24–hour rolling deletion; failure reverts to scheduled full–on mode. Near–term pilot: the southern 1 km of the main spine.

S5 Memory–Collection Vehicle

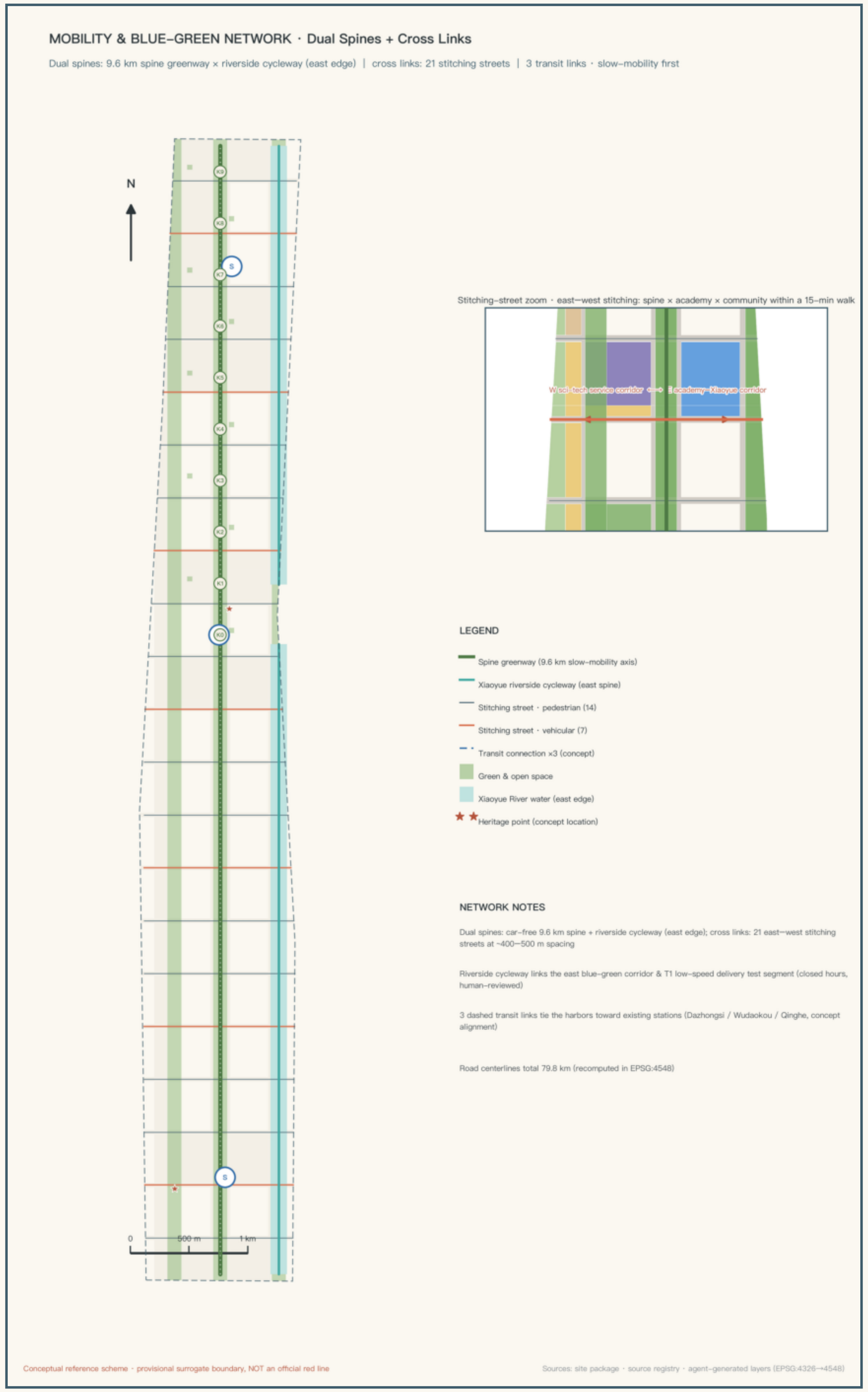
Appointment — on–site collection — narrator confirmation — authorized archiving — optional display; AI only transcribes and pre–screens; every archive is manually checked, signed, and withdrawable. Near–term pilot: 2—3 communities in the South Harbor.

T3 Robot Staging Unit

Application — safety assessment — staging–area public notice and testing — public feedback — formalization or exit; test reports are public; accidents terminate the test and restore the site to public space. Near–term pilot: North Harbor modules M1—M3.

Privacy boundary: no facial recognition anywhere in the belt; personal data processed on–device by default; every scenario has explicit data boundaries, human review, and exit mechanisms.

Blue–Green Network · Slow Mobility and City Landmarks



Blue–Green Skeleton & Slow Mobility

The main–spine park belt (a continuous green corridor approx. 120 m wide) and the Xiaoyue River blue–green corridor on the eastern edge run side by side north—south as the 'dual spines'; the twenty–one stitching streets form east—west public links; with ten K–node pocket parks and twenty–one platform–park squares they build a three–tier 'avenue—park—square' public–space system. Computed green coverage: approx. 29.5% (standalone layer) / 29.7% (land–balance caliber).
The main–spine greenway runs approx. 9.6 km linking all ten K0—K9 station markers (nine segments, ten stations); the Xiaoyue River waterfront cycling path is the second north—south slow–mobility line on the east; the twenty–one 'stitching streets' cross east—west at approx. 500 m intervals (seven double as secondary vehicular roads, the rest are slow–mobility only).

AI Pilgrimage Landmarks (one primary landmark + three public memory & contribution nodes · all conc

Primary Landmark · 'New Bell' Light–and–Sound Installation

South Harbor: a real–time dialogue between the 600–year Yongle Great Bell's voiceprint and the AI urban soundscape — the core of the belt's auditory identity and its only primary landmark.

① Kilometer–Zero Origin Square · Public Memory Node

K0 · Central Harbor: the narrative kilometer zero of the belt (a conceptual numbering, not the historical mileage origin of the railway); paving scored on the 1,435 mm standard gauge — an urban square for staying and gathering.

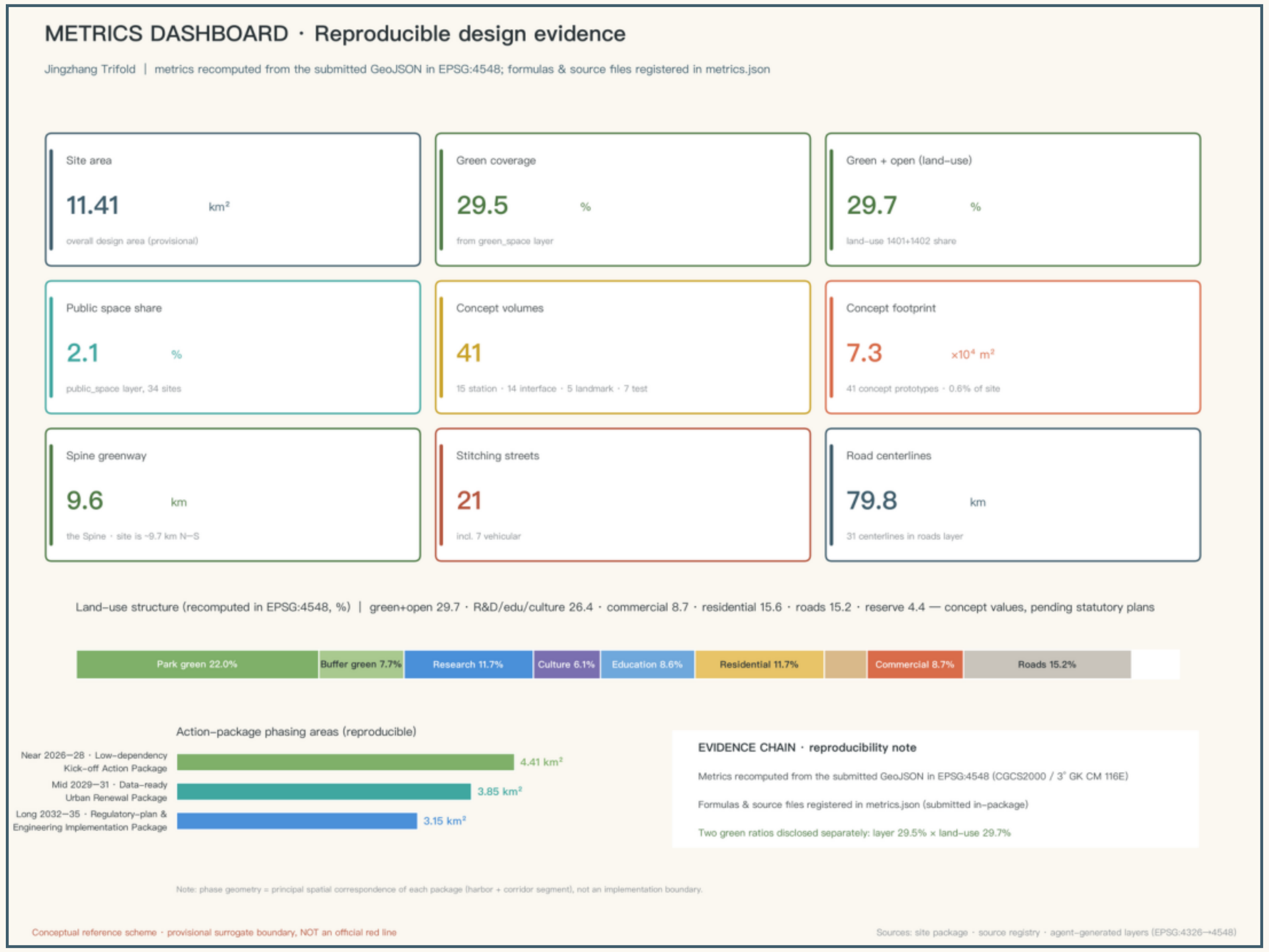
② Herringbone Switchback Bridge & Memory Hall · Public Memory Node

Near K2: accessible ramps and a 'problem—experiment—failure—correction' exhibition make the switchback an experienceable mechanism; the hall beneath shows oral histories and railway relics of the old Qinghuayuan station.

③ K9 Gate of the Future · Public Memory Node

Northern end of the North Harbor: a walkable, lingerable, humble threshold signaling 'the next corridor' northward with low–brightness light pulses at night — not a high–recognition sculptural landmark.

Metrics · Area Recalculation and Phasing



Core Metrics (computed values · provisional caliber · reproducible under EPSG:4548)

Computed site area
11.4128 M sq m
Announcement caliber approx. 11.40 M sq m; deviation approx. 0.1%

Three key areas (computed)
192.9 / 104.3 / 72.0 ha
Deviations from announcement values all under 0.5%

Green coverage
29.5% / 29.7%
Standalone green–layer caliber / land–balance caliber

Public–space ratio
≈ 2.1%
Union of 34 standalone squares, approx. 238,900 sq m

Concept massing prototypes
41 sites · 73,400 sq m
Footprint ratio approx. 0.6% — flagship–node intent only, not existing buildings

Main spine / stitching streets
9.6 km / 21 streets
Site length approx. 9.7 km N–S; road centerlines total approx. 79.8 km

Phasing Plan (evidence gates + action packages · conceptual recommendation)

Evidence–gate logic: each action package starts only after its prerequisite evidence is in place — the near–term package needs no official boundary or ledger; the medium–term package gates on property–rights, existing–building and municipal ledgers; the long–term package gates on regulatory–plan conditions and engineering demonstration.

Near term 2026—2028 · Low–dependency Kick–off Action Package
approx. 4.412 M sq m

Actions that need no official boundary or ledger: brand & wayfinding pilots, oral–history collection, 'Four Seasons of the Trifold' events, movable facilities, micro–renewal of negotiable South Harbor interfaces

Medium term 2029—2031 · Data–ready Urban Renewal Package
approx. 3.853 M sq m

Once property–rights, existing–building, road, and municipal data are in place: three–harbor community renovation, stitching–street repairs, learning–vein corridor renewal

Long term 2032—2035 · Regulatory–plan and Engineering Implementation Package
approx. 3.148 M sq m

Facilities requiring regulatory–plan conditions and engineering demonstration: Herringbone Switchback Bridge, Intelligence Meridian utility corridor, large public buildings, North Harbor blank–land activation

Statutory control indicators (FAR, building height, coverage ratio, green–ratio control values, setbacks) are uniformly recorded as 'pending official data' until official Regulatory Detailed Planning conditions are supplied; this proposal draws no statutory planning conclusions. Recalculation triggers are registered in metrics.json and assumptions.json. The three phase areas describe the principal spatial correspondence ranges of the action packages (harbors and their radiating corridor segments) and do not constitute implementation boundaries.

Evidence · Sources and Compliance Statements

Sources (used per public-registry grading)

① Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources: Pre-Qualification Announcement (2026-05-09)

② User-provided & rights-cleared: Taskbook excerpts for the open call to global agents (2026-05-18)

③ Beijing Municipal S&T Commission & Zhongguancun Administrative Committee: 'Three Zones and Two Wings' (2026-04-03)

④ Ministry of Education: 'The Eight Colleges on Beijing's Xueyuan Road' (2009-08-24)

⑤ Beijing Daily: verification of the railway's historical kilometer zero (2026-08-11); 'Three Zones and Two Wings' layout report

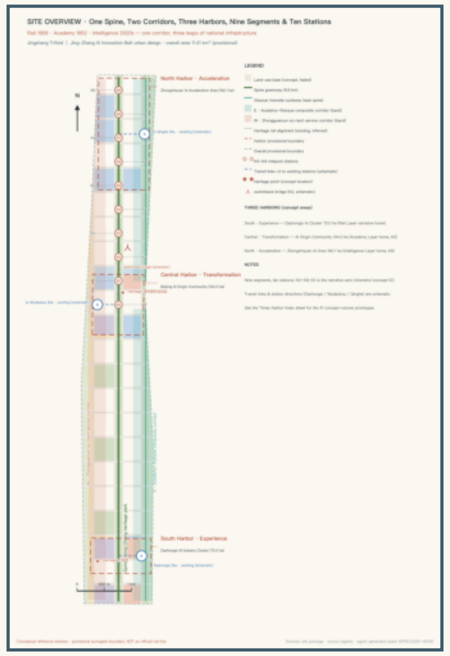
⑥ China Railway official website: Jing-Zhang Railway cultural-heritage materials (herringbone switchback at Qinglongqiao)

⑦ MOHURD Urban Design Management Measures (2017); MNR Land & Sea Use Classification Guide (2023)

⑧ Barrier-Free Environment Construction Law (2023); CAC et al.: Interim Measures for Generative AI Services (2023)

⑨ Haidian converged media: AI Origin Community clustering-scale report (2026-07-29, publicity-caliber background data)

⑩ Repository registry: provisional coarse boundary & three key-area polygons + public source registry



Provisional Boundary Statement

As of submission, the organizer has not released the official precise boundary or key-area polygons. The site boundary and three key areas adopt the repository-registered provisional coarse boundary, used only for AI generation, visualization, and local self-checking; it must not be interpreted as an official planning boundary, an approval basis, or a basis for precise area figures. Once official polygons are published, land use, buildings, roads, green space, public space, phasing, and all area metrics must be recomputed.

Metric Reproducibility

All area-type metrics are computed as polygon areas from the submitted GeoJSON under EPSG:4548 (CGCS2000 three-degree zone, central meridian 117°E); ratio-type metrics use the computed provisional-boundary area as denominator; any third party can reproduce them with the same method. The machine-audit layer is fully preserved in sources.json · metrics.json · compliance_matrix.json · standard_matrix.json · design_depth_matrix.json · geometry/*.geojson.

Boundary of Conceptual Recommendations

This proposal represents no government review, approval, or implementation commitment; all spatial recommendations are conceptual; the provisional boundary is not a red line; FAR, height, retain-renovate-demolish decisions, alignments, capacities, investment, and scheduling are all non-final; no testing scenario constitutes approved operation; statements on funding, policy, and investment attraction are development directions and constitute no government commitment.

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